

Human and LLM Facilitation in Collaborative Design Meetings: A Computational Behavioral Comparison

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Abstract

Large language models (LLMs) are increasingly being explored as facilitators of group discussion and collaborative work, but the behavioral differences between LLM-generated and human facilitation remain insufficiently characterized. This study presents a context-matched computational comparison of human Project Manager facilitation turns from scenario-based collaborative design meetings and Llama 3.1 8B Instruct responses across 199 paired contexts from the AMI Meeting Corpus. The analysis covers five confirmatory dimensions—contextual semantic distance, a lexical elaboration proxy, empathy-and-hedge lexical density, topic divergence, and phase-vocabulary alignment—plus an exploratory directiveness measure that failed domain validation and is not interpreted. Human facilitation was higher on contextual semantic distance ($r = 0.644$) and on empathy-and-hedge lexical density ($r = 0.203$), a surface proxy dominated by hedging rather than psychological empathy. Llama-generated facilitation was higher on topic divergence ($r = 0.703$), the lexical elaboration proxy ($r = 0.288$; length-confounded), and instruction-conditioned phase-vocabulary alignment ($r = 0.382$). A secondary no-phase check preserved the Llama-higher phase contrast and did not establish spontaneous phase awareness. The comparisons do not support a claim of facilitation superiority. The resulting behavioral profile provides a computational reference point for future research comparing LLM facilitators, human facilitators, prompts, models, and collaborative settings.

Keywords: LLM facilitation; large language models; human-AI collaboration; human-LLM comparison; meeting facilitation; collaborative design; conversational AI; behavioral analysis; design meetings; AMI Meeting Corpus.

1 Introduction

Large language models (LLMs) are increasingly being explored as facilitators of group discussion, collaboration, decision making, and design [Korre et al., 2025, Dong et al., 2024, Alsobay et al., 2026]. Work on LLM facilitation has begun to study facilitation systems and group outcomes. A complementary question is what the facilitation language itself looks like. This paper studies

how LLM-generated facilitation differs behaviorally from human facilitation in collaborative design meetings.

Existing research has examined how LLM-based facilitation can be implemented, when LLMs should intervene, and how facilitation affects group interaction or outcomes. A complementary question is how the linguistic behavior of LLM facilitators differs from naturally occurring human facilitation when both are considered in comparable conversational contexts. We address this question through a context-matched computational behavioral comparison of human and LLM facilitation in collaborative design meetings.

The comparison uses scenario meetings from the AMI Meeting Corpus [Carletta et al., 2006] and Llama 3.1 8B Instruct [Grattafiori et al., 2024]. We characterize the behavioral profile of LLM-generated facilitation against human Project Manager turns on five confirmatory dimensions: contextual semantic distance, a lexical elaboration proxy, empathy-and-hedge lexical density, topic divergence, and phase-vocabulary alignment. Directiveness is exploratory and unvalidated. The profile is a description of facilitation language, not a judgment of facilitation quality.

The research questions are:

1. What behavioral differences distinguish LLM-generated facilitation from human facilitation in collaborative design meetings?
2. Which measurable behavioral dimensions most strongly differentiate LLM-generated and human facilitation turns?
3. How robust are these behavioral differences to decoding temperature and sensitivity to representation, response length, duplicate contexts, and phase prompting?

The contributions are as follows.

1. A context-matched behavioral comparison of human and LLM facilitation using 199 paired contexts.
2. A multi-dimensional computational characterization spanning contextual semantic distance, lexical elaboration, hedge-and-empathy lexical density, topic divergence, and phase-vocabulary alignment. These measures are not presented as a validated universal taxonomy.
3. Robustness analysis across response length, duplicate contexts, LDA topic count, TF-IDF representation, decoding temperature, phase-label removal, and mixed-effects modeling.
4. A behavioral reference point for future research on LLM facilitation. The study does not establish a universal benchmark.

2 Literature Review

2.1 LLM facilitation systems

Korre *et al.* [Korre et al., 2025] survey how LLMs are being used to evaluate and facilitate online discussions, including intervention strategies and open research directions. Dong, Ding, and Ito [Dong et al., 2024] designed a multi-phase facilitation agent that uses GPT-3.5 to produce facilitation messages for large-scale online discussion. That work is a system design: it specifies when a phase-based agent should speak and how an LLM is controlled. It does not compare the wording of those messages with human facilitation in matched conversational contexts.

2.2 Human and LLM facilitation behavior

Tsirmpas, Korre, and Pavlopoulos [Tsirmpas et al., 2026] studied a different behavioral question: when a facilitator should intervene. In their survey of online-discussion excerpts, human experts were more cautious about intervening, whereas the LLMs they tested were more eager to facilitate. Their comparison concerns the decision to intervene, not the linguistic form of a facilitation turn written for the same preceding context. Neighboring studies of human-LLM co-writing [Lee et al., 2022] and story writing [Yuan et al., 2022] likewise characterize collaborative language generation, but not meeting facilitation.

2.3 Group outcomes

Alsobay *et al.* [Alsobay et al., 2026] ran a preregistered experiment in which live groups received no facilitation, a one-time prompt, a human facilitator, or a GPT-4o facilitator. LLM facilitation increased information sharing by raising the minimum level of task engagement. Neither human nor LLM facilitation significantly changed the final group decision. That result is about downstream participation and decision outcomes. It does not describe the computational properties of the facilitation language itself.

2.4 Where this study fits

Rather than evaluating a facilitation system solely by intervention timing or downstream group outcomes, this study characterizes the behavioral properties of the facilitation language itself. The comparison uses context-matched human and LLM facilitation turns from scenario-based collaborative design meetings. In that setting, Starostka *et al.* [Starostka et al., 2021] distinguished a method approach to design-thinking facilitation from a cofacilitation approach, and Seidel and Fixson [Seidel and Fixson, 2013] studied how novice teams use needfinding, brainstorming, and prototyping. Brown [Brown, 2008] described design work as movement through inspiration, ideation, and implementation rather than a fixed sequence of steps. Protocol studies of design communication [Stempfle and Badke-Schaub, 2002, Kan and Gero, 2017] and computational displays of spoken meetings [Chandrasegaran et al., 2019] show that meeting language can be measured, but they do not supply a human-LLM facilitation comparison. Yu *et al.* [Yu et al., 2025] discuss opportunities and reliability limits of generative AI in design, including hallucination. Gmeiner *et al.* [Gmeiner et al., 2023] studied how designers learn to co-create with AI manufacturing tools. Neither paper measures facilitation turns. The computational measures used here—sentence embeddings [Reimers and Gurevych, 2019], latent Dirichlet allocation [Blei et al., 2003], and a surface lexical pipeline—are instruments for that comparison. They are not a judgment of facilitation quality.

3 Method

3.1 Corpus and data processing

The analyzed meetings are elicited scenario meetings from the AMI Meeting Corpus [Carletta et al., 2006, Carletta, 2006]. They are not a sample of generic workplace meetings. In the scenario, four participants—a project manager (PM), an industrial designer, a user-interface designer, and a marketing expert—take a project through four meetings in a day [Carletta, 2006]. The design task is a new remote control [AMI Project, 2006]. The corpus release also contains other meetings; this comparison uses the scenario design meetings and the project-manager turns within them.

We used the text transcription subset comprising 279 transcript files. Transcripts were segmented into sentence-level speaker turns, yielding 89,472 sentence-turns across 279 sessions. In the AMI scenario, the project manager runs the meetings [Carletta, 2006], and those turns are the source of the human facilitation sample. Session phases were assigned by relative position in each transcript (first 20% $\rightarrow$ Empathize/Define; middle 50% $\rightarrow$ Ideate; final 30% $\rightarrow$ Prototype/Evaluate). These labels are an author-defined temporal heuristic. They are not a published phase model, and they are not ground-truth phase annotation.

Candidate facilitation moves were extracted from Project Manager turns with rule-based filters: (a) word count between 8 and 60 words, (b) absence of casual non-meeting fragments, (c) presence of facilitation signals (interrogative syntax, facilitation keywords such as “what if” or “how might,” or modal verbs in turns of at least 12 words), and (d) co-occurrence with design-meeting vocabulary. This process identified 2,685 candidate facilitation moves across 139 sessions. Of those candidates, 152 were labeled Ideate and 2,533 Prototype/Evaluate. None fell in the first temporal window. The evaluated set contains 199 of these candidates: 149 Ideate and 50 Prototype/Evaluate. The released preprocessing script reconstructs that set from the context identifiers stored with the generated responses. It does not record an additional random draw from the candidate pool. The evaluated human sample therefore represents rule-selected candidate facilitation moves rather than an exhaustive or independently validated annotation of all PM facilitation behavior. A 60-turn sheet for facilitation-move agreement was prepared, but the ratings were left blank, so no inter-rater coefficient is available.

3.2 LLM inference setup

Matched LLM facilitation responses were generated using Meta Llama 3.1 8B Instruct [Grattafiori et al., 2024]. Preprocessing, feature extraction, and all statistical analyses were run locally on a laptop with an NVIDIA GeForce RTX 3050 Ti (4 GB VRAM). That GPU cannot hold Llama 3.1 8B, so response generation was the step run on Google Colab, in 4-bit format via `bitsandbytes` (NF4 quantization, double quantization, float16 compute dtype) on an NVIDIA Tesla T4 (16 GB VRAM).

For each of the 199 human contexts, a prompt was constructed containing a system prompt defining the facilitator role (a single move of 1–3 sentences) and a user prompt containing the preceding 5 conversational turns (truncated to a maximum of 180 words), the explicit design-phase label, and the generation request. Context matching refers to the same extracted conversational window supplied to the LLM, not to equal informational access between the LLM and the human PM during the original meeting. Responses were generated at three temperatures ($T \in \{0.3, 0.7, 1.0\}$) with `do_sample=True`, `top_p=0.9`, and `max_new_tokens=150`, producing 597 valid LLM responses. A random seed for this primary generation is not recorded in the released generation script.

The primary LLM prompt explicitly includes the temporal phase label (for example, “Current design phase: Ideate”). A secondary generation repeated the same 199 contexts, model, 4-bit settings, temperatures, and sampling options with that line removed (seed = 42; all 597 responses passed the existing validity filter). Phase-vocabulary alignment was rescored with the original token-proportion function. This contrast is a sensitivity analysis of the phase-label confound and is not added to the five-dimension confirmatory family. The system prompt still asks for a design-facilitation move, so a no-phase response is not an uninstructed turn.

3.3 Behavioral dimensions and operationalization

Each facilitation move was evaluated on six computational dimensions. Five are confirmatory. Directiveness is exploratory.

1. **Contextual semantic distance (F_1).** $F_1 = 1 - \cos(\mathbf{e}_{\text{ctx}}, \mathbf{e}_{\text{resp}})$, with embeddings from all-MiniLM-L6-v2. Cosine comparison of sentence embeddings follows Sentence-BERT [Reimers and Gurevych, 2019]. Higher values indicate greater embedding-space distance from the preceding context.
2. **Directiveness (exploratory / unvalidated).** Probability $P(\text{directive})$ from a DistilBERT classifier [Sanh et al., 2019] trained on 197 synthetic examples (99 directive, 98 open-ended). Human domain validation on 100 AMI utterances yielded Cohen’s $\kappa = -0.0231$, indicating chance-level agreement. This measure is excluded from confirmatory inference.
3. **Lexical elaboration proxy (F_3).** Composite of named-entity density, type-token ratio, and normalized average word length. Named-entity density uses spaCy [Honnibal et al., 2020]. Normalized word length is $\min(\text{AWL}/10, 1)$. This metric captures surface lexical elaboration. The robustness analysis investigates word-count confounding.
4. **Empathy-and-hedge lexical density (F_4).** Proportion of empathy and hedging terms relative to turn word count. This is a surface lexical proxy. It does not measure psychological empathy. Median scores are zero for both groups.
5. **Topic divergence (F_5).** Jensen–Shannon distance between context and response topic distributions from a 15-topic LDA model [Blei et al., 2003] trained jointly on all human and LLM texts. The number of topics was selected heuristically. Sensitivity to that choice is reported below.
6. **Phase-vocabulary alignment (F_6).** Proportion of response tokens matching predefined phase-specific vocabulary lists. The primary score uses responses generated with the phase label in the prompt. The no-label generation is a secondary sensitivity analysis. It is not a measure of spontaneous phase awareness.

3.4 Statistical analysis

The 199 human PM turns and 199 LLM responses at $T = 0.7$ form matched pairs by context identifier. The primary test is the paired Wilcoxon signed-rank test [Wilcoxon, 1945]. Bonferroni correction is applied once across the five confirmatory dimensions. A test is rejected when its raw p -value is below $\alpha/5 = 0.010$, which is the same decision as rejecting when the adjusted value $p_{\text{Bonf}} = \min(1, 5p)$ is below 0.05. Comparing p_{Bonf} with 0.010 would apply the correction twice. Effect sizes are reported as $r = |Z|/\sqrt{n}$ from the normal approximation [Rosenthal, 1994], with 95% bootstrap confidence intervals (2,000 resamples, seed = 42). Directiveness is excluded from Bonferroni correction and treated as exploratory.

Two paired Wilcoxon tests at $T = 0.7$ —human versus no-phase, and label-present versus no-phase—share a separate Bonferroni correction over those two tests. They do not change the significance status of the five confirmatory dimensions. Temperature-specific label contrasts are Bonferroni-adjusted across the three temperatures and are not added to either family as confirmatory tests.

The 597 LLM responses are 199 contexts crossed with three temperatures. The inferential temperature test is a Friedman test within context, with Bonferroni correction over the five

confirmatory dimensions. A Spearman correlation on the stacked 597 rows is reported only as a description, because that correlation does not model the repeated contexts.

The robustness checks are as follows. (a) Length-matched subsampling keeps pairs whose relative word-count difference is below 0.50, which leaves $n = 73$ pairs, and repeats the paired Wilcoxon test. The relative difference is $|WC_{\text{human}} - WC_{\text{LLM}}| / (WC_{\text{human}} + 1)$. (b) A Spearman correlation relates each paired difference to human word count. (c) A linear mixed-effects model uses an LLM indicator as the fixed effect and the context identifier as a random intercept.

4 Results

4.1 Overall behavioral profile

Human and LLM facilitation turns differed across the five confirmatory computational dimensions (Figure 1; Table 3). At $T = 0.7$, human Project Manager turns were higher on contextual semantic distance and on empathy-and-hedge lexical density. Llama 3.1 8B Instruct responses were higher on topic divergence, the lexical elaboration proxy, and phase-vocabulary alignment. All five contrasts met the primary Bonferroni rule: raw $p < 0.010$, equivalently $p_{\text{Bonf}} < 0.05$. Table 1 states what each contrast supports and what it does not. LLM responses were longer (mean 38.5 words at $T = 0.7$) than human turns (mean 23.5 words; Table 2), which matters for the length-sensitive scores.

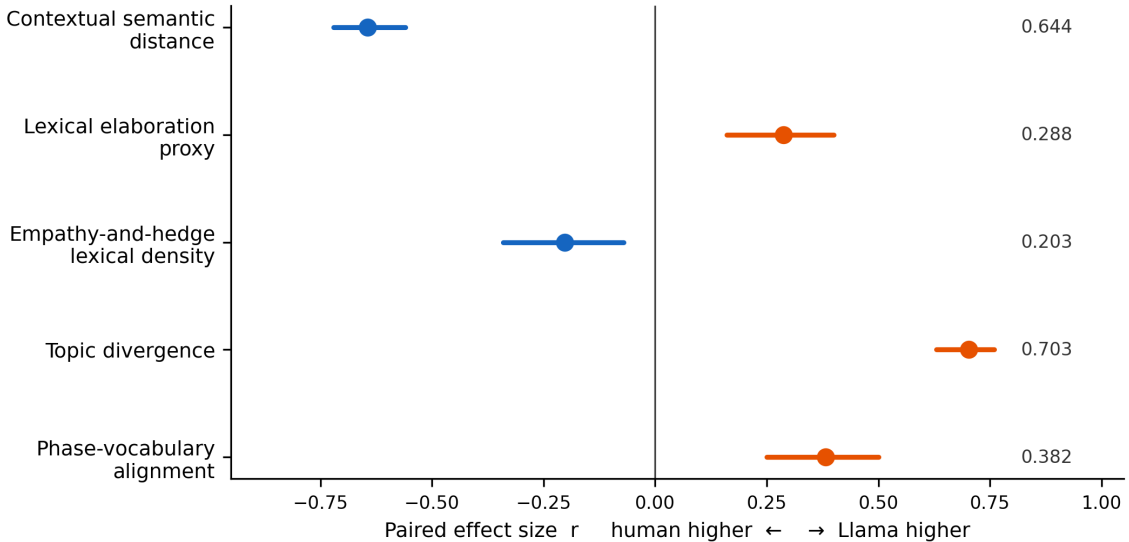


Figure 1: Primary behavioral profile at $T = 0.7$ ($n = 199$ pairs). Points are the paired effect size r , signed so that negative values mean the human turn is higher and positive values mean the Llama response is higher. Intervals are the 95% bootstrap confidence intervals on r , given the same sign. Directiveness is omitted because it is unvalidated.

Taken together, the results indicate that LLM and human facilitators exhibit distinguishable behavioral profiles even when responding to the same conversational contexts. The primary family is these five tests only. The secondary phase-label family is corrected separately and is not part of this decision. The exploratory directiveness row in Table 3 is not interpreted.

Table 1: What the five confirmatory contrasts support. Directions are human versus Llama 3.1 8B Instruct at $T = 0.7$.

Dimension	Observed difference	What it supports	What it does not establish
Contextual semantic distance	Human higher	Greater distance in <code>all-MiniLM-L6-v2</code> space	General semantic superiority
Lexical elaboration	Llama higher	Greater surface lexical elaboration	Conceptual specificity
Empathy/hedging	Human higher	A lexical hedge/empathy difference	Psychological empathy
Topic divergence	Llama higher	Greater LDA topic-distribution divergence	Creativity
Phase alignment	Llama higher	Greater phase-vocabulary alignment	Spontaneous phase awareness

Table 2: Descriptive statistics for human PM turns and Llama 3.1 8B Instruct responses. Types are unique lowercased word tokens. Mean sentences count terminal punctuation.

Source	N	Words	SD	Types	Sents
Human PM	199	23.5	12.7	725	1.5
LLM $T = 0.3$	199	40.5	7.9	1031	1.8
LLM $T = 0.7$	199	38.5	7.5	1125	1.6
LLM $T = 1.0$	199	37.6	8.5	1196	1.5

Table 3: Paired Wilcoxon signed-rank tests: human PM turns versus Llama 3.1 8B Instruct ($T = 0.7$, $n = 199$ pairs). $p_{\text{Bonf}} = \min(1, 5p_{\text{raw}})$ for the five confirmatory dimensions and is compared with 0.05 (equivalently, raw p with 0.010). Effect size $r = |Z|/\sqrt{n}$. The 95% confidence interval is a bootstrap with 2,000 resamples.

Dimension	H Mdn	L Mdn	r	95% CI	p_{Bonf}	Sig.
Contextual distance	0.742	0.593	0.644	[0.56, 0.72]	<0.001	***
Lexical elaboration	0.397	0.416	0.288	[0.16, 0.40]	<0.001	***
Empathy/hedge density	0.000	0.000	0.203	[0.07, 0.34]	0.021	*
Topic divergence	0.540	0.744	0.703	[0.63, 0.76]	<0.001	***
Phase-vocabulary alignment ^a	0.000	0.000	0.382	[0.25, 0.50]	<0.001	***
Directiveness (unvalidated) ^b	0.021	0.036	0.026	[0.00, 0.17]	0.718	ns

^aThe primary prompt included the phase label. A secondary generation without it is in Table 4. ^bExcluded from confirmatory inference: domain $\kappa = -0.02$. Stars use the adjusted p -value: one star means $p_{\text{Bonf}} < 0.05$, and three stars mean $p_{\text{Bonf}} < 0.001$. The empathy raw p -value is 0.00415.

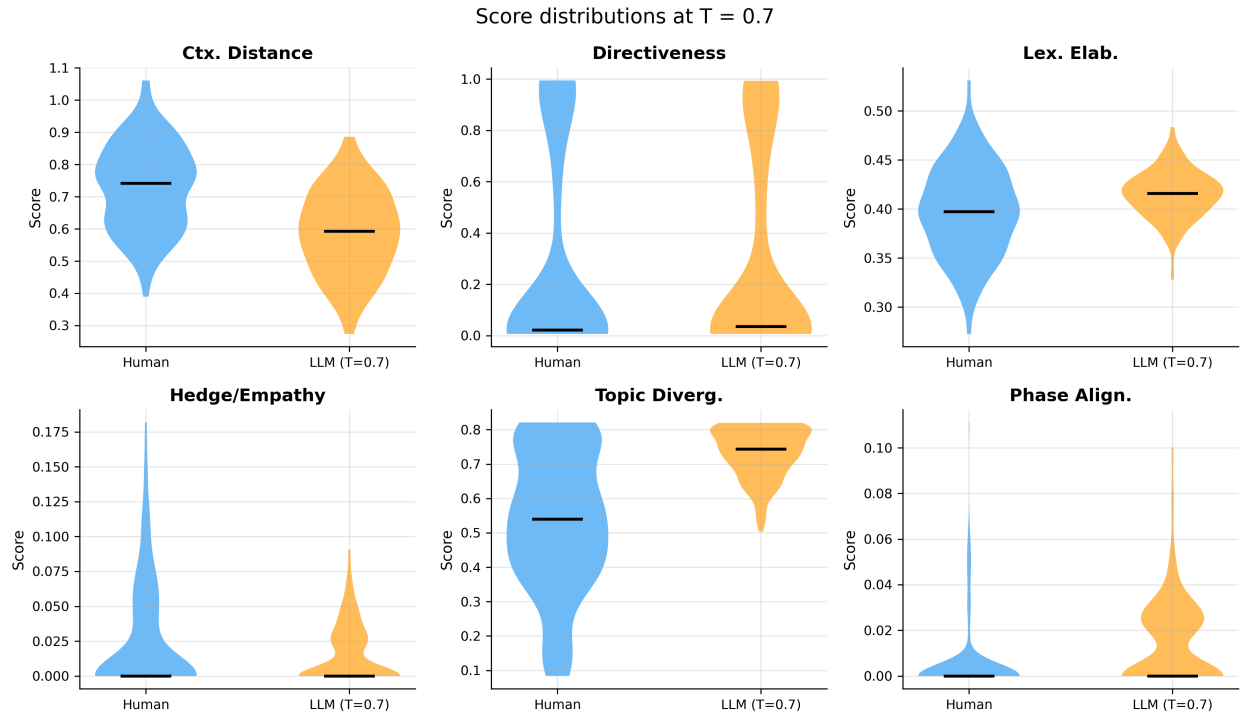


Figure 2: Score distributions for human facilitation and Llama 3.1 8B Instruct at $T = 0.7$. Horizontal lines are medians. The hedge/empathy panel is lexical density, not a psychological rating. Directiveness is exploratory and unvalidated.

4.2 Contextual semantic distance

Human PM facilitation turns showed greater contextual distance from the preceding conversational context than Llama 3.1 8B Instruct responses in `all-MiniLM-L6-v2` embedding space ($r = 0.644$, 95% CI [0.56, 0.72]). TF-IDF cosine distance did not reproduce the contrast ($r = 0.002$, $p = 0.978$). The result is representation-specific. It is not a general claim that human facilitation was more semantically diverse.

4.3 Topic divergence

Llama 3.1 8B Instruct responses showed greater divergence from the preceding context in LDA topic-distribution space than human PM facilitation turns ($r = 0.703$, 95% CI [0.63, 0.76]). The same Llama-higher direction held when the topic model was refit at $K = 5, 10, 15, 20$, and 30 (paired $r = 0.715, 0.655, 0.598, 0.599$, and 0.445). This is divergence between fitted topic distributions. It does not establish creativity or innovation, and it is a different quantity from the TF-IDF distance.

4.4 Lexical elaboration proxy

LLM-generated facilitation turns showed higher lexical elaboration than human facilitation turns ($r = 0.288$, 95% CI [0.16, 0.40]). The paired difference is substantially associated with human word count (Spearman $\rho = -0.596$, $p < 0.001$). The measure is surface lexical elaboration and is length-confounded. It does not establish conceptual specificity, informativeness, or explanation quality.

4.5 Empathy and hedging

Human facilitation turns showed higher hedge-and-empathy lexical density than Llama-generated facilitation ($r = 0.203$, 95% CI [0.07, 0.34]). The raw p -value is 0.00415, which is below 0.010. Equivalently, $p_{\text{Bonf}} = \min(1, 5p) = 0.021$, which is below 0.05. Under that single five-test rule the contrast is significant, and it is the weakest confirmatory effect. Both medians are zero. Human turns contain 3 empathy-only dictionary hits and 116 hedge hits; Llama responses at $T = 0.7$ contain 30 empathy-only hits and 94 hedge hits. The signal is dominated by hedges. The measure does not establish psychological empathy.

4.6 Phase-vocabulary alignment

In the primary label-present generation, Llama responses showed greater phase-vocabulary alignment than human turns ($r = 0.382$, 95% CI [0.25, 0.50]). Both medians are zero (means 0.0050 and 0.0149). This primary result is instruction-conditioned phase-vocabulary alignment. Figure 3 prints the raw means on separate scales so the smaller empathy and phase means remain visible.

4.7 Secondary phase-label sensitivity

The no-phase generation is a secondary check, not a sixth confirmatory dimension. Rescoring the stored label-present responses reproduces the published phase means (human 0.0050; Llama $T = 0.7$ with the label 0.0149). Without the label the $T = 0.7$ mean is 0.0115 (Figure 4). All three medians are zero. Two tests at $T = 0.7$ share one Bonferroni correction, separate from the five-dimension family (Table 4).

Removing the explicit phase label did not eliminate the Llama-higher phase-vocabulary alignment contrast ($r = 0.294$, 95% CI [0.16, 0.42], raw $p = 3.3 \times 10^{-5}$, secondary $p = 6.6 \times 10^{-5}$). The mean

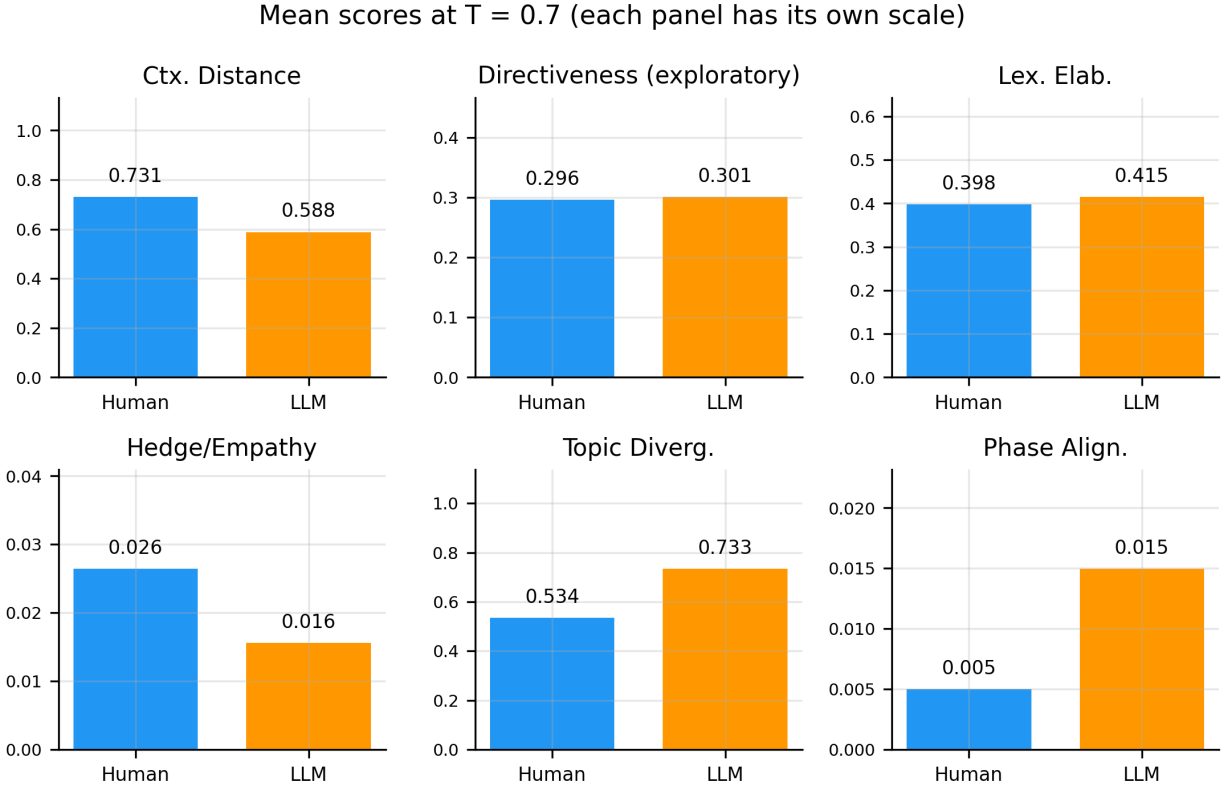


Figure 3: Mean scores for human facilitation and Llama responses at $T = 0.7$. Each panel starts at zero and uses its own scale. Printed numbers are the raw means. The hedge/empathy panel is lexical density. Directiveness is exploratory and unvalidated.

score was somewhat lower without the label, while the label-present versus no-label difference was not statistically significant after correction ($r = 0.123$, 95% CI [0.01, 0.26], raw $p = 0.082$, secondary $p = 0.163$). The data therefore do not establish that explicitly naming the phase amplifies the score. The same label contrast is nonsignificant at $T = 0.3$ and $T = 1.0$ after correction across the three temperatures (raw $p = 0.042$ and 0.036). Because the system prompt still instructed the model to produce a design-facilitation move, the no-phase analysis does not establish spontaneous phase awareness. A nonsignificant label contrast is not evidence that the label has no effect.

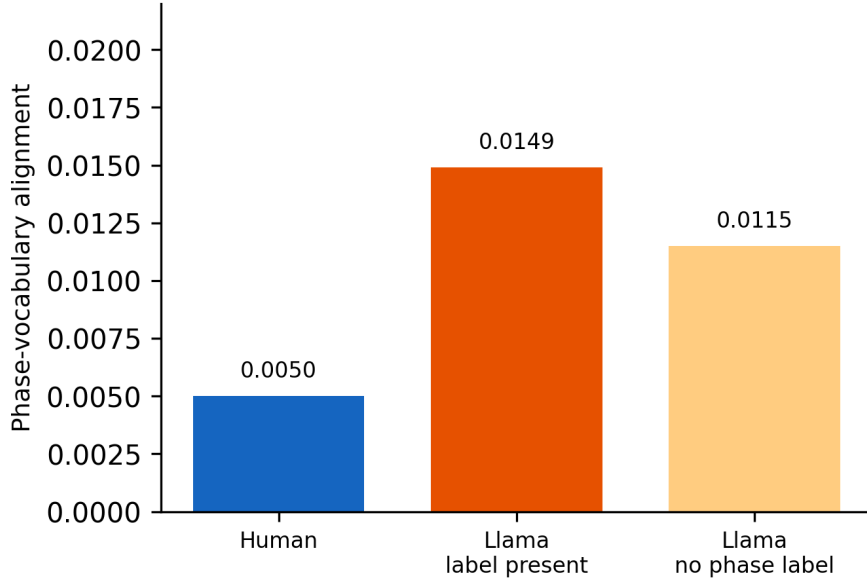


Figure 4: Phase-vocabulary means at $T = 0.7$. Removing the explicit phase label did not eliminate the Llama-higher contrast. The label-present versus no-label difference was not statistically significant after correction. The system prompt still requested a design-facilitation move.

Table 4: Secondary phase-label sensitivity at $T = 0.7$ ($n = 199$ matched contexts). The two raw p -values are Bonferroni-adjusted together. This family is separate from the five confirmatory dimensions. Effect size $r = |Z|/\sqrt{n}$, with the same 2,000-resample bootstrap as the primary tests.

Contrast	Means	r	95% CI	Raw p	Sec. p
Human vs. no label	0.0050 vs. 0.0115	0.294	[0.16, 0.42]	<0.001	<0.001
Label vs. no label	0.0149 vs. 0.0115	0.123	[0.01, 0.26]	0.082	0.163

4.8 Robustness and temperature

The checks below answer alternative explanations of the behavioral profile. They are not a second primary family.

Within-context Friedman tests found no statistically detectable temperature effect on the confirmatory dimensions across $T \in \{0.3, 0.7, 1.0\}$ after Bonferroni correction (Table 5). A nonsignificant result does not establish that temperature has zero effect. Descriptive temperature correlations are in Appendix A. They stack repeated contexts and are not the inferential result.

Table 5: Friedman tests of scores across $T \in \{0.3, 0.7, 1.0\}$ within the same 199 contexts. Bonferroni correction uses the five confirmatory dimensions.

Dimension	χ^2	p	p_{Bonf}	Sig.
Contextual distance	0.44	0.802	1.000	—
Lexical elaboration	4.04	0.133	0.664	—
Empathy/hedge density	2.21	0.332	1.000	—
Topic divergence	5.46	0.065	0.327	—
Phase-vocabulary alignment	5.61	0.060	0.302	—
Directiveness (exploratory)	0.85	0.654	0.654	—

Length-matched pairs ($n = 73$) kept the same direction for contextual distance ($r = 0.514$), lexical elaboration ($r = 0.807$), topic divergence ($r = 0.712$), and phase alignment ($r = 0.487$). Response length substantially accounts for the elaboration contrast, which is why that proxy stays a surface measure. Empathy-and-hedge density in that subset ($r = 0.231$, $p = 0.048$) does not meet the primary raw threshold of 0.010. That nonsignificant subset test is not evidence that the groups are equivalent. Removing 63 verbatim duplicate contexts leaves 136 pairs. The five confirmatory directions remain and still meet the same Bonferroni rule (contextual distance $r = 0.678$, lexical elaboration $r = 0.326$, empathy-and-hedge $r = 0.276$, raw $p = 0.001$, $p_{\text{Bonf}} = 0.006$, topic divergence $r = 0.712$, phase alignment $r = 0.318$). Repetition does not by itself produce the directional pattern. Topic divergence stays Llama-higher from $K = 5$ to $K = 30$, so the direction is not an artifact of $K = 15$ alone. TF-IDF does not reproduce the embedding-distance result, so that contrast is not generic semantic distance. Mixed-effects estimates have the same sign as the paired differences on all five confirmatory dimensions (Appendix A).

5 Discussion

5.1 A behavioral profile of LLM facilitation

Context-matched LLM and human facilitation turns exhibit distinguishable behavioral profiles even when responding to the same conversational contexts. Those differences describe generated language. They do not establish facilitation quality.

5.2 Contextual alignment and divergence

Human PM facilitation turns exhibited greater contextual distance from the preceding conversational context than Llama 3.1 8B Instruct responses in **all-MiniLM-L6-v2** embedding space ($r = 0.644$). Llama 3.1 8B Instruct responses exhibited greater divergence from the preceding context in LDA topic-distribution space, with the direction remaining stable across topic counts from 5 to 30. The two contrasts are not interchangeable: TF-IDF cosine distance did not reproduce the embedding result ($r = 0.002$, $p = 0.978$). The embedding and topic patterns occur together in 126 of 199 pairs (63%). For human-AI interaction, the practical point is that distance from context depends on the representation. For computational social science, the same paired design can carry more than one distributional measure without treating them as one construct.

5.3 Surface lexical behavior

LLM-generated facilitation turns showed higher lexical elaboration, although this measure was substantially associated with response length (Spearman $\rho = -0.596$, $p < 0.001$) and should therefore be interpreted as a surface lexical proxy rather than conceptual specificity. Length matching changed the size of that effect ($r = 0.807$ at $n = 73$) without removing the direction. Future prompting studies that change response length should expect this proxy to move with length.

5.4 Human hedge/empathy language

Human facilitation turns showed higher hedge-and-empathy lexical density ($r = 0.203$, raw $p = 0.00415$, $p_{\text{Bonf}} = 0.021$), although the measure was dominated by hedge terms and does not establish psychological empathy. Both medians are zero. The length-matched subset does not meet the primary raw threshold ($p = 0.048$). That result is not evidence of equivalence.

5.5 Phase-conditioned facilitation language

The Llama-higher phase-vocabulary alignment contrast persisted when the explicit phase label was removed ($r = 0.294$), although this sensitivity analysis did not establish spontaneous phase awareness. The label-present versus no-label comparison was not significant after its own correction, so the data do not establish that printing the phase name amplifies the score. The system prompt still asked for a design-facilitation move. Full-session lengths are not stored, so other temporal boundaries were not re-estimated, and the evaluated set contains 149 Ideate turns and 50 Prototype/Evaluate turns.

5.6 Implications for future LLM-facilitation research

The measures and context-matched design provide a behavioral reference point against which future facilitation studies can be compared. The same paired setup can be repeated for other LLMs, other prompts, other domains, and other meeting structures. This study does not itself establish those generalizations, and it does not replace work on facilitation systems [Dong et al., 2024], intervention timing [Tsirmpas et al., 2026], or group outcomes [Alsobay et al., 2026]. For HCI, the profile is evidence about human–AI facilitation behavior, limited to measurable language differences. For NLP and computational social science, it is a computational characterization of conversational behavior, including the split between embeddings, TF–IDF, and LDA. For design research, it characterizes facilitation language in collaborative design meetings. For LLM research, the comparison describes the behavioral profile observed when an LLM is assigned a facilitation role. That profile does not show that the model recognizes design phases.

Directiveness remains unresolved ($\kappa = -0.0231$). No conclusion about human versus LLM directiveness is warranted.

The reference point is bounded by one model, Llama 3.1 8B Instruct in 4-bit; human turns from project managers in the AMI remote-control scenario rather than from professional facilitators at large; author-defined phase bins; a length-confounded elaboration proxy; a hedge-dominated empathy lexicon; an embedding result that TF–IDF does not reproduce; topic divergence defined on a heuristically chosen 15-topic model even though the direction is stable at other K ; 63 duplicated contexts, with the directional pattern still present in the 136 unique pairs; and a paired test that does not by itself model session clustering, which the mixed-effects check only partly addresses.

6 Conclusion

This study provides a context-matched computational characterization of LLM and human facilitation in collaborative design meetings. Human facilitation is higher on embedding distance and on hedge-and-empathy lexical density. Llama 3.1 8B Instruct is higher on topic divergence, surface lexical elaboration, and instruction-conditioned phase-vocabulary alignment. The no-phase test leaves the Llama-higher phase contrast in place and does not establish spontaneous phase awareness. These findings characterize measurable properties of facilitation language rather than facilitation quality, creativity, psychological empathy, or general intelligence. The behavioral profile reported here provides a reference point for future research comparing LLM facilitators across models, prompts, domains, and collaborative settings.

Data Availability

The AMI Meeting Corpus is distributed by its maintainers at <https://groups.inf.ed.ac.uk/ami/corpus/>. This paper does not redistribute that corpus. The analysis code, derived feature tables, and generated facilitation responses used in the comparison are available at <https://github.com/Ayaan577/llm-facilitation-behavior>.

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A Additional displays and robustness

Pairwise Spearman correlations among the six scores remain weak ($|\rho| < 0.28$; Figure 5). An exploratory principal component analysis of all 796 turns attributes 22.8% of variance to PC1 and 21.3% to PC2 (44.1% together; Figure 6). Human turns lie toward negative PC1 and LLM responses toward positive PC1. This decomposition is descriptive. It is not a validated facilitation-style construct. Figure 7 shows descriptive associations with temperature. The inferential temperature tests are the Friedman tests in Table 5.

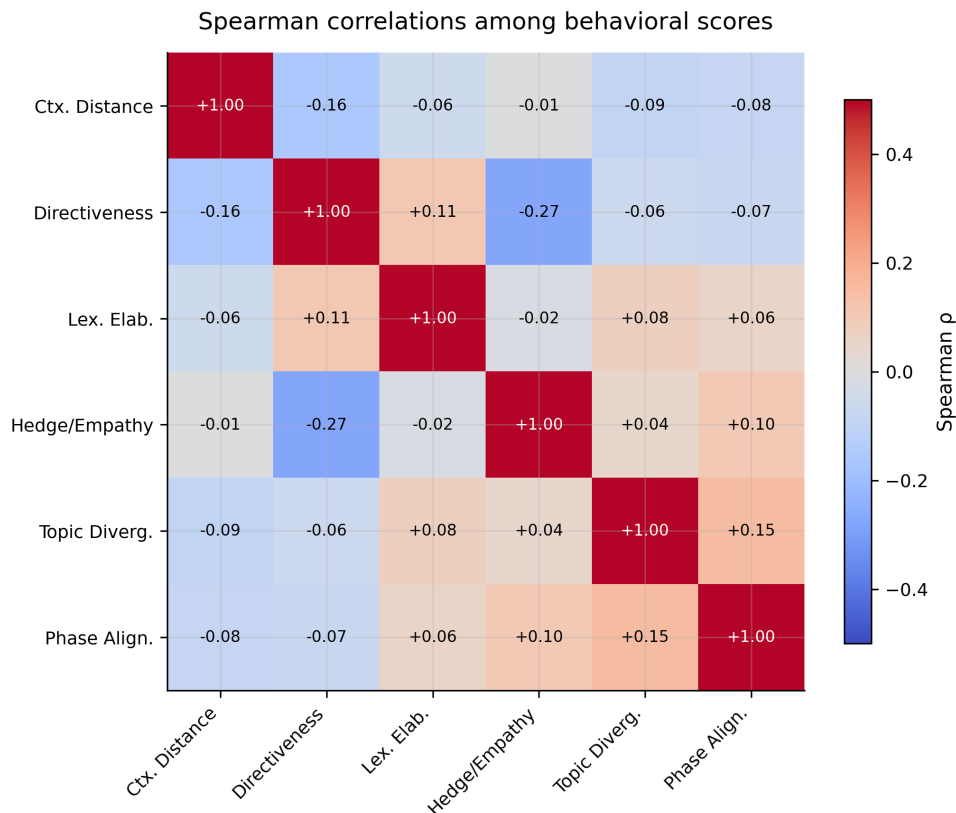


Figure 5: Spearman correlations among the six scores ($N = 796$), including the unvalidated directiveness score.

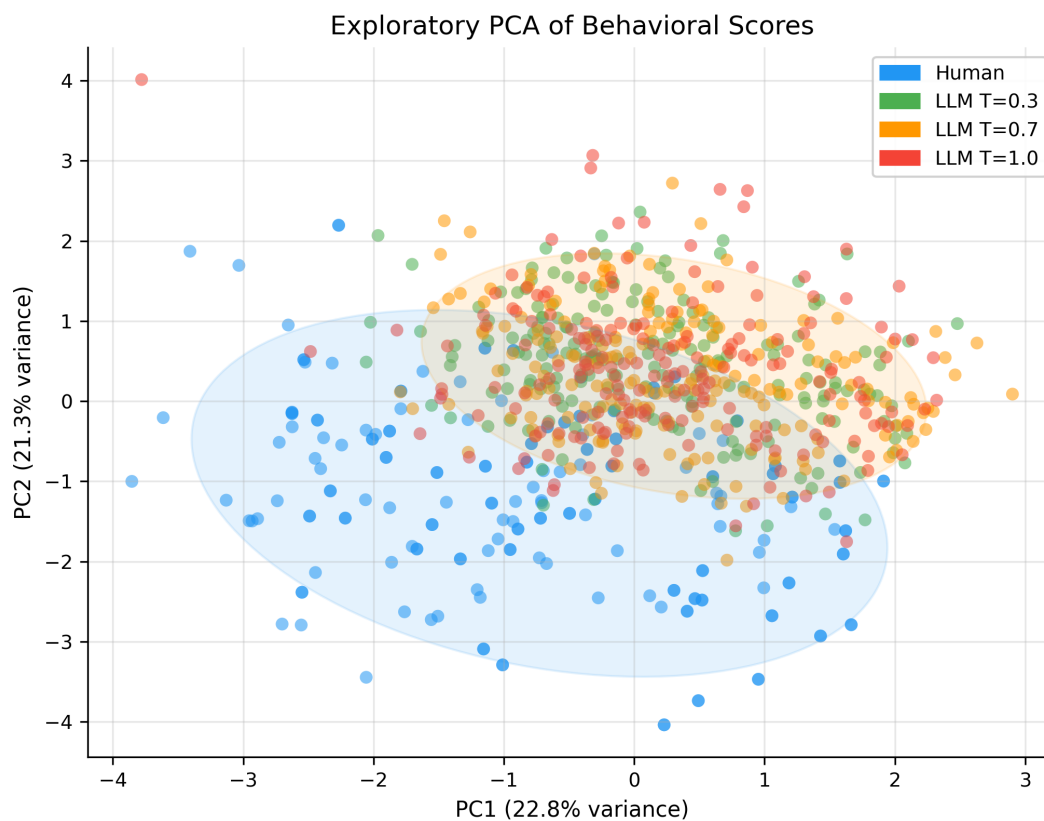


Figure 6: Exploratory PCA of behavioral scores for human turns and Llama responses at all three temperatures, fit on all 796 turns. The axes are principal components of the score matrix, not a validated style space.

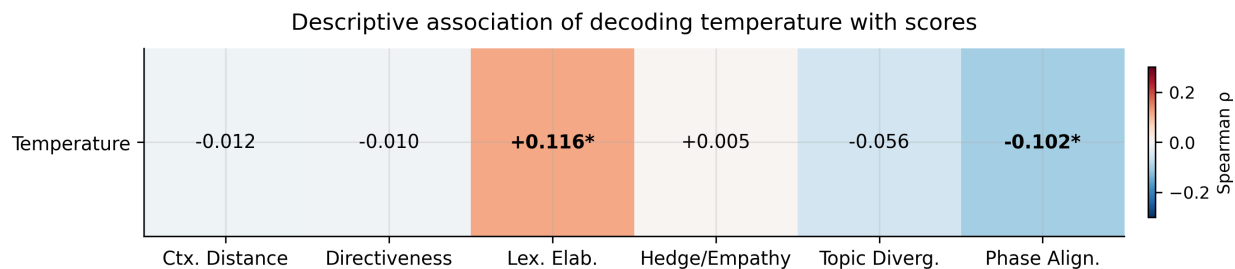


Figure 7: Descriptive Spearman correlations between decoding temperature and scores on 597 stacked responses. These are not the within-context Friedman tests.

Table 6: Robustness checks: length-matched pairs ($n = 73$), Spearman correlation of the paired difference with human word count, and a mixed-effects estimate with a random intercept for context. Primary r is from the paired Wilcoxon test ($n = 199$).

Dimension	Prim. r	Len. r	Part. ρ	ME β	p_{Bonf}	Note
Contextual distance	0.644	0.514	-0.137	-0.143	<0.001	Stable
Topic divergence	0.703	0.712	-0.027	+0.199	<0.001	Stable
Lexical elaboration	0.288	0.807	-0.596	+0.017	<0.001	Length-confounded
Phase alignment ^a	0.382	0.487	-0.059	+0.010	<0.001	Instruction-conditioned
Empathy/hedge density	0.203	0.231	+0.068	-0.011	0.021	Hedge proxy
Directiveness (exploratory)	0.026	—	—	+0.005	0.718	Unvalidated

^aThe primary prompt included the phase label. The separate no-label comparison did not show a significant label effect after correction.